

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (CL)

November 2012

CL308 Geotechnical Engineering-II

Date: 07.11.2012, Wednesday

Time: 10:00a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a)** Enlist the methods for boring the holes in to soil. Explain rotary drilling and core drilling in detail. [05]
- (b)** List out the equipments used in Standard Penetration Test. [02]
- Q - 2 (a)** Draw different shapes of wells along with the factors affecting the choice of particular shape. [06]
- (b)** Write the assumptions made in Rankine's theory and give the formulae for of active earth pressure for cohesionless soil for the given case (a) dry backfill with zero surcharge (b) submerged backfill on one side and (c) submerged backfill on two sides. [05]
- (c)** Write short note on well foundation & show its component parts in neat sketch. [03]

OR

- Q - 2 (a)** What is open caisson? Explain in detail the sinking process of the well foundation. [06]
- (b)** Differentiate between active earth pressure & passive earth pressure. [05]
- (c)** Write short note on grip length. [03]
- Q - 3 (a)** A load of 1200 kN acts as a point load at the surface of a soil mass. Determine the stress at a point 4 m below and 3 m away from the point of action of load by using (a) Boussinesq's formula & (b) Westergaard's theory ($m=0$). [05]
- (b)** A soil mass is retained by a smooth backed vertical wall of 6.0 m height. The soil has bulk unit weight of 20 kN/m^3 and $\phi = 16^\circ$. The top of the soil is level with the top of the wall and it is horizontal. If the soil surface carries a UDL of 4.5 kN/m^2 , determine the total active thrust on the wall per meter of wall and its point of application. [05]
- (c)** List out the assumptions for Boussinesq's theory for stress distribution in soil. [04]

OR

- Q - 3 (a)** A concentrated load of 50 kN acts on the surface of soil. Find out the variation of vertical stress increment due to load on the horizontal plane at a depth of 2 m up to a

horizontal distance of 5 m on either side of centre and plot the same on graph paper.

- (b) For the retaining wall shown in **figure-1**, assume the active condition. Use Rankine's active earth pressure theory to determine; [05]
- (i) The Active force on the retaining wall, and
 - (ii) The location of the resultant line if cohesion is zero.
- (c) What is Newmark's Influence Chart? Explain the procedure to use it. [04]

SECTION – II

- Q - 4 (a) Explain finite slopes and infinite slope with example. [02]
- (b) Enlist the types of failures of the slope. [02]
- (c) Which are the types of factor of safety used in stability analysis of slopes? [03]
- Q - 5 (a) Differentiate between end bearing pile and frictional pile. [03]
- (b) A group of 9 piles, 9 m long is used as the foundation of column. The piles are 30 cm in diameter with centre to centre spacing 90 cm. the subsoil consists of clay with unconfined compressive strength = 170 kN/m^2 . Estimate safe load. Take F.O.S. = 2. [06]
- (c) Explain the Swedish Circle method for the analysis a cohesive frictional soil. [05]

OR

- Q - 5 (a) Enlist the situations where pile foundation is suitable. [03]
- (b) A reinforced concrete pile of 30 cm X 30 cm and 10 m long is driven into coarse sand extending to a great depth. The average unit weight of sand is 18 kN/m^3 and average N-value is 15. Determine the allowable load on pile making use of N-value. F.O.S.=4 [06]
- (c) An embankment is constructed using clayey sand. The properties of soil are as following: Cohesion= 25 kN/m^2 , Angle of internal friction= 22° and Unit weight= 20 kN/m^3 . The height of embankment is 8 m, $\alpha=30^\circ$ & $\beta=20^\circ$. Determine the factor of safety. Use method of slices and considering toe failure. [05]
- Q - 6 (a) A 3 m square footing is located in dense sand at a depth of 2 m. the moist unit weight of sand above the water table is 18 kN/m^3 and the saturated unit weight is 20 kN/m^3 . The shear strength parameters are $c=0$ and $\phi = 35^\circ$. Determine the ultimate bearing capacity for the following cases: (a) water table is at G.L. (b) water table at footing level. Take $N_q=33$; $N_\gamma=34$. [07]
- (b) Differentiate between general shear failure and local shear failure. [05]
- (c) Explain negative skin friction. [02]

OR

- Q - 6 (a) What will be the gross and net safe bearing capacity of sand having $\phi = 36^\circ$ and effective unit weight is 18 kN/m^3 under the following cases; [07]

- (i) 1 m wide strip footing and (ii) 1m X 1m square footing.

Consider the footings are placed at a depth of 1 m from ground surface and water table is at a great depth. Take F.O.S. =3; $N_q=47$; $N_\gamma=43$. Use Terzaghi's theory.

- (b) List out the assumptions for Terzaghi's bearing capacity theory. [05]
 (c) Write formulae for static analysis of pile driven in sand and clay. [02]

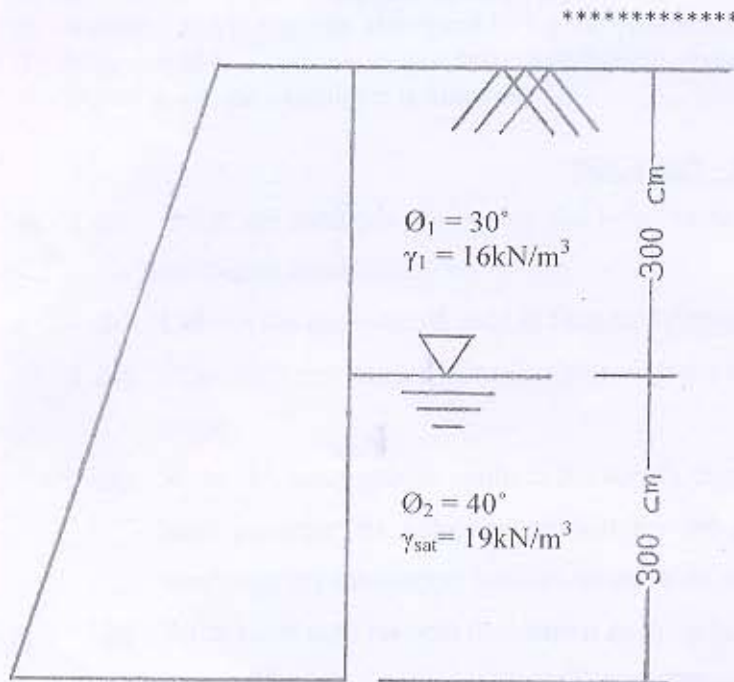


Figure-1 : Q-3 (b)